

Cross validation

DJ



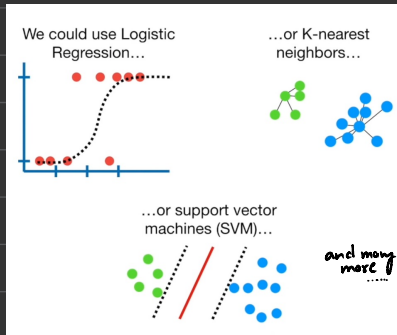
Cross Validation

Chest Pain	Good Blood Circ.	Blocked Arteries	Weight	Heart Disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
...	...	...	...	...

Suppose we have this dataset and wants to predict for some new datapoint.

Chest Pain	Good Blood Circ.	Blocked Arteries	Weight	Heart Disease
Yes	No	No	168	???

1st we want to decide which ML algo is best for this.



Cross validation allows us to use diff ML algo. and get a sense of how well they will work in practice

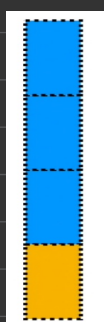
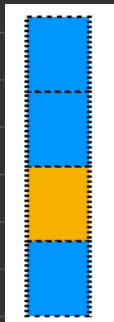
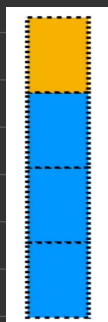
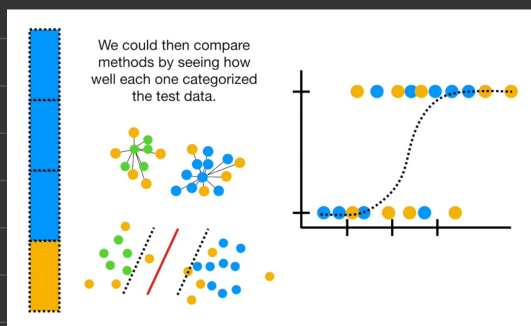
We need to perform two things

1. Estimate parameters
[in ML lingo this is known as training the algo]
2. Evaluate ML method
[testing]



Dividing the data into
test and train.

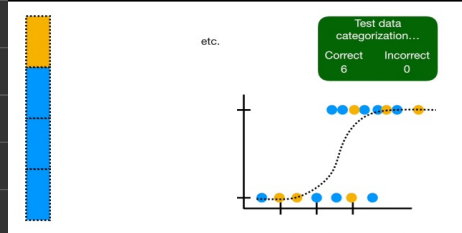
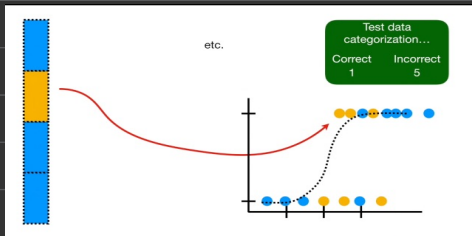
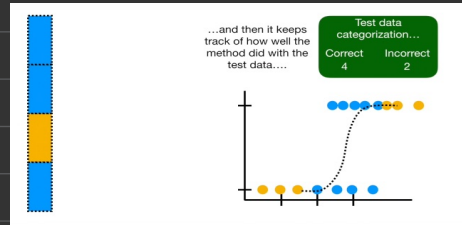
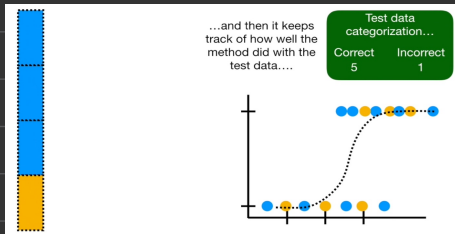
Suppose 75% train
25% test



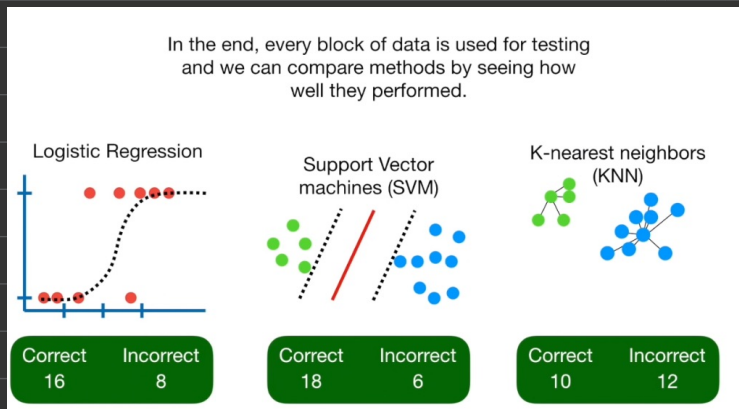
→ We can use either
1st, 2nd, 3rd, 4th
block for testing.

But which block will
give me better
results??

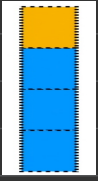
Cross validation uses all cases
one by one and summarizes the
result at end.



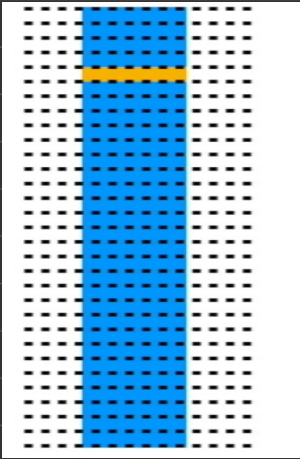
will compare above all results



In this case SVM did the best job so we will use that.



When we divide our dataset into 4 parts that is known as Four Fold Cross validation



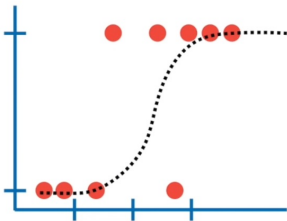
In an extreme case, we would call each individual of a block.

This is called **Leave one Out Cross Validation**

Each sample tested individually.

* In practice it is very common to use $CV=10$

Say like we wanted to use a method that involved a **"tuning parameter"** - a parameter that isn't estimated, but just sort of guessed. (For example, Ridge Regression, has a tuning parameter)...



→ Here we can use 10-fold cross val. to help find the best value for that tuning parameter.

